



Band-SCNet: A Causal, Lightweight Model for High-Performance Real-Time Music Source Separation

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Abstract

Music source separation (MSS) for real-time applications faces challenges due to significant performance loss. In this paper, we propose Band-SCNet, a lightweight real-time model that bridges the performance gap between real-time and non-real-time models. Band-SCNet combines Sparse Compression with Cross-band and Narrow-band Blocks to reduce model size and complexity while improving performance. We also introduce the Compressed Self-Attention (CSA) Fusion Module, which enhances efficiency by reducing parameters. Experiments on the MUSDB18-HQ dataset show that Band-SCNet outperforms existing real-time models with an SDR of 7.79 dB, 2.59 million parameters by relaxing to a 92 ms latency, confirming its suitability for real-time MSS. The model achieves a balance between performance, latency, and model size, offering a promising solution for real-time music source separation.

Index Terms: Music Source Separation, Real-time, Lightweight

1. Introduction

Music source separation (MSS) is a process that isolates individual music sources from a mixed audio signal, such as vocals, drums, bass, and other instruments [1]. The growing demand for real-time, lightweight MSS models has become particularly important as these models are needed to operate on a variety of devices in streaming applications. Such models can provide key services like specific track extraction, personalized music rendering, and improved listening experiences. However, despite this growing need, there is a limited amount of research focused on developing real-time, lightweight MSS solutions that can effectively balance performance and computational efficiency.

The existing ones are based on the U-Net structure, comprising the encoder, decoder, and bottleneck. Wave-U-Net [2] proposes a real-time time-domain model that uses masking for latent domain and contrastive learning, with an encoder that compresses embedding dimension and a decoder that performs reconstruction, with Dual-Path RNN (DPRNN) [3] in the bottleneck. HS-TasNet [4], a real-time low-latency model, integrates frequency- and time-domain. It contains two encoder for compression, one for each domain. It uses LSTMs in the bottleneck to fuse the output of encoders. For lightweight models, DTTNet [5] uses a frequency-domain model consists of Time-Frequency Convolutions Time-Distributed Fully-connected UNet (TFC-TDF UNet) [6], with downsampling in the encoder and upsampling in the decoder. Its bottleneck integrates the TFC-TDF Modules with an improved DPRNN by

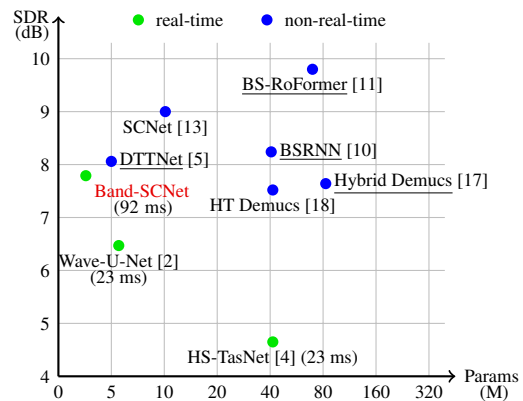


Figure 1: Performance and model size of MSS models. The numbers in parentheses represent latency. Underlined models are single-output models. Our proposed Band-SCNet bridges the gap between real-time and non-real-time models while maintaining a lightweight size.

splitting the channels. However, it is a single-output model, requiring four separate models to generate four instruments.

In the non-real-time MSS research, most methods are frequency-domain methods [7, 8, 9, 10, 11, 12, 13], with some being time-domain methods [14, 15] or the fusion of time-domain and frequency-domain methods [16, 17, 18]. Due to the high sampling rate of music signal (e.g. 44.1 kHz), most of these models have limited performance or high complexity as shown in Figure 1, making them unsuitable for real-time applications. SCNet [13], a frequency-domain model, proposes the Sparse Compression method, which reduces complexity while maintaining significant performance. This makes SCNet a promising candidate for real-time adaptation.

Real-time and non-real-time models exhibit a significant performance gap (Figure 1). Music contains both spectral (e.g. pitch) and temporal information (e.g. beat). However, real-time models struggle with incomplete temporal information, as future data is unavailable. Therefore, how to better utilize the spectral information is a challenge for real-time models, especially under the constraint of lightweight. Existing real-time models, Wave-U-Net and HS-TasNet, are limited because: 1) they treat low- and high-frequency bands equally during compression, ignoring the fact that low-frequency bands contain more information; 2) the 23 ms window length used in these real-time MSS models is too short for accurately capturing the full dynamics of low-frequency components of music signals; 3) they learn the full spectrogram as a whole, rather than at multiple scales, making it challenging to capture the intricate

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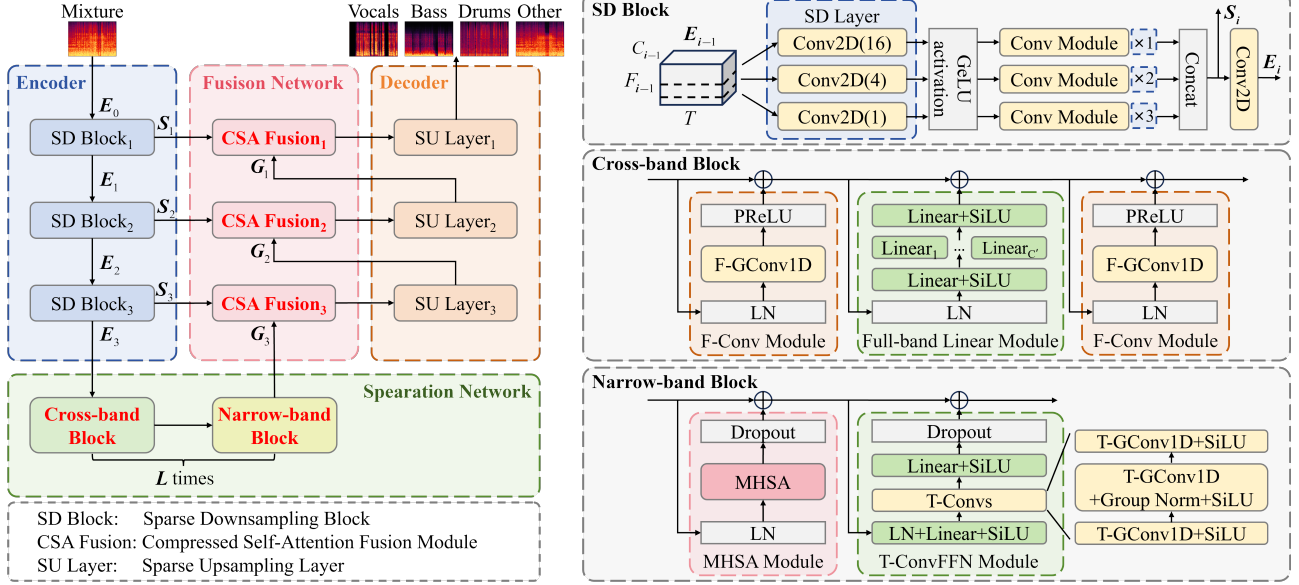


Figure 2: Structure and building blocks of Band-SCNet based on Online SCNet, with modifications including Cross-band Block, Narrow-band Block, and CSA Fusion Module.

details and variations at different time-frequency resolutions.

The first issue can be addressed by Sparse Compression method, as proposed by SCNet. For the second issue, adopting longer window lengths improve frequency resolution, enhancing the representation of stationary components for more accurate source separation. This adjustment strikes a balance between real-time performance and separation quality, allowing the model to capture richer temporal information without greatly increasing latency. Recent research in speech separation has demonstrated significant performance of the full-band and sub-band modeling approach [19]. SpatialNet [20] leverages interleaved Cross-band and Narrow-band Blocks to effectively capture cross-band and narrow-band spectral information. This method can address the third issue mentioned above. However, this approach has not been explored in MSS due to the higher sampling rate and longer window length for music result in a large frequency dimension, leading to high complexity. Sparse Compression provides a natural solution to this issue.

In this paper, we propose a real-time lightweight model Band-SCNet. Our contributions are as follows:

- We implement the real-time version of SCNet, called Online SCNet, as our baseline.
- We propose Band-SCNet, a lightweight real-time model that combines Sparse Compression with Cross-band and Narrow-band Blocks to enhance performance, reduce model size, and maintain acceptable complexity.
- We introduce the CSA Fusion Module, which integrates Multi-Head Self-Attention (MHSA) [21] with a linear gate and Gated Linear Unit (GLU) [22] to improve performance while reducing the number of parameters.
- Experiments show that Band-SCNet achieves an SDR of 7.79 dB, surpassing state-of-the-art real-time models with only 2.59 million parameters on the MUSDB18-HQ dataset [23].

2. Online SCNet

SCNet contains an Encoder, a Separation Network and a Decoder with Fusion Modules. Encoder compresses the frequency

dimension and enhances the feature dimension. Separation Network captures the temporal and the spectral information. Decoder reconstructs and outputs four spectrograms with the help of Fusion Network.

Detailed structure of Encoder and Decoder is illustrated in section 3.2. Its Separation Network consists of several DPRNN layers, with Fourier Transforms between them as feature conversion, overcoming diminishing returns from stacking more layers. Fusion Module is composed of 2D convolutional layer (Conv2D) and GLU.

To turn SCNet into a real-time model, we replace all convolutions with causal convolutions, replace Layer Normalization (LN) [24] with cumulative layer normalization (cLN) [25], and replace the Bidirectional LSTMs along the time dimension in DPRNN with Unidirectional LSTMs. We also remove the Fourier Transforms, as it operates along the whole sequence and cannot be used in real-time models. This modification also reduces the number of parameters.

3. Band-SCNet

3.1. Overall structure

Band-SCNet consists of the same Encoder and Decoder as Online SCNet, with additional modifications such as Cross-band Blocks and Narrow-band Blocks in Separation Network, and the CSA Fusion Module in Fusion Network (Figure 2). These additions enable Band-SCNet to more effectively capture spectral information while maintaining a smaller model size and an acceptable latency for real-time applications.

Short-Time Fourier Transform (STFT) is applied to convert the waveform $x \in \mathbb{R}^{M \times N}$ into spectrogram and stack the real and imaginary parts, getting the input $X \in \mathbb{R}^{2M \times T \times F}$. M, N, T, F denote the number of channels, samples, frames and frequency bins, respectively.

3.2. Encoder and Decoder - Sparse Compression

Encoder is composed of Sparse Downsampling (SD) Blocks (Figure 2). SD Block contains a SD Layer and several Con-

volution Modules. SD Layer is designed to make the model place more emphasis on low-frequency information compared to high-frequency information. SD Layer divides the frequency dimension into low, medium, and high parts, and performs downsampling by convolution with stride of 1, 4, and 16 respectively. This process is called Sparse Compression. During compression, SD Layer also increases the feature dimension. Several Convolution Modules are applied after SD Layer. To effectively capture low-frequency information, SD Block includes 3 Convolution Modules at low-frequency part, 2 at medium-frequency part, and 1 at high-frequency part. SD Block_{*i*} takes $E_{i-1} \in \mathbb{R}^{C_{i-1} \times T \times F_{i-1}}$ as input, where C denotes the feature dimension and i denotes the index of SD Block, and finally outputs the compressed features $E_i \in \mathbb{R}^{C_i \times T \times F_i}$ and the skip connection $S_i \in \mathbb{R}^{C_i \times T \times F_i}$.

Decoder is composed of Sparse Upsampling (SU) Layers, performing upsampling by deconvolutions with strides of 1, 4, and 16 for the low-, medium-, and high-frequency parts.

3.3. Separation Network

The Separation Network comprises L groups of alternating Cross-band and Narrow-band Blocks, adapted from Spatial-Net [20]. It takes E_3 as input and produces $G_3 \in \mathbb{R}^{C_3 \times T \times F_3}$ as output.

Cross-band Block To capture the cross-band information, Cross-band Block integrates two Frequency-Convolutional (F-Conv) Modules and one Full-band Linear Module, as shown in Figure 2. F-Conv Module is designed to capture correlations between neighboring frequencies, which contains a LN layer, a grouped convolution layer along the frequency dimension (F-GConv1D) and a parametric rectified linear unit (PReLU) activation function [26]. To learn full-band information, Full-band Linear Module is applied. In this module, a linear layer followed by the sigmoid-weighted linear unit (SiLU) activation function [27] is first used to reduce the feature dimension from C_3 to C' . Then we use a group of linear layers along the frequency dimension. Different feature dimensions use different linear layers, denoted as Linear_i for $i = 1, \dots, C'$. Finally, we apply a linear layer and a SiLU activation function to increase the feature dimension back to C_3 .

Narrow-band Block To capture the narrow-band information, Narrow-band Block consists of MHSA Module and Time-Convolutional Feed Forward Network (T-ConvFFN) Module, which is illustrated in Figure 2. In order to capture the narrow-band information, MHSA Module is applied, which is composed of a LN layer, a MHSA layer and a dropout. MHSA layer processes each frequency independently. T-ConvFFN Module is a modified version of feed forward network used in Transformer [21]. In the T-ConvFFN Module, a linear layer followed by a SiLU activation function is used to increase the feature dimension from C_3 to C'' , to capture more complex features. Then a Time-Convolutional (T-Convs) Module is applied, which consist of three grouped convolution layers along the time dimension (T-GConv1D) followed by a SiLU activation function. An additional Grouped Normalization (GN) [28] layer is added after the second T-GConv1D. Finally, the feature dimension is reduced back to C_3 by a linear layer followed by a SiLU activation function and a dropout.

3.4. Fusion Network - CSA Fusion Module

To enhance spectrogram reconstruction and further improve performance, we introduce CSA Fusion Module. This module comprises a Compressed Multi-Head Self-Attention (CMHSA)

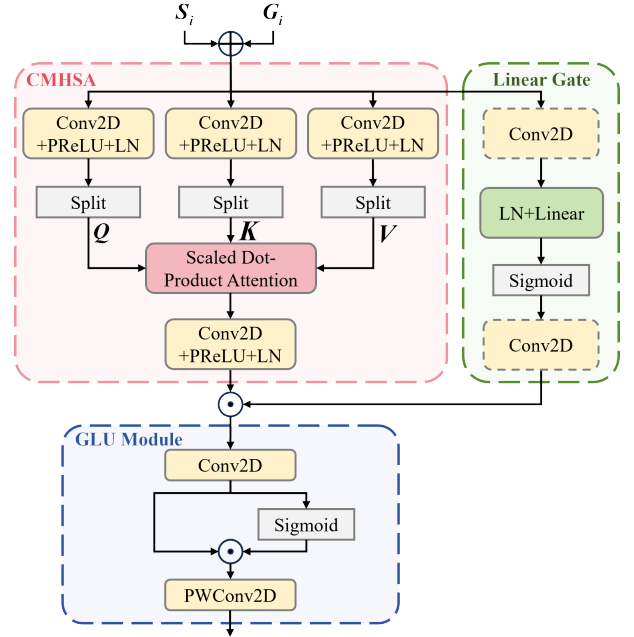


Figure 3: CSA Fusion Module

Module, a Linear Gate, and a GLU Module (Figure 3).

The objective of CMHSA Module is to learn the global correlations of S_i and G_i , to better reconstruct the spectrogram. First, S_i and G_i are summed and then passed through three Conv2D followed by a PReLU activation function and a LN layer. Inspired by TF-CrossNet [29], this process generates the query $Q \in \mathbb{R}^{(C_d \times H) \times T \times F_i}$, the key $K \in \mathbb{R}^{(C_d \times H) \times T \times F_i}$ and the value $V \in \mathbb{R}^{(C_i/H \times H) \times T \times F_i}$, where $C_d = \lceil C_a/F_i \rceil$. C_a represents the embedding dimension when computing the attention, and H denotes the number of heads. Then we split and reshape them to $Q_h \in \mathbb{R}^{T \times (C_d \times F_i)}$, $K_h \in \mathbb{R}^{T \times (C_d \times F_i)}$ and $V_h \in \mathbb{R}^{T \times (C_i/H \times H) \times T \times F_i}$, where h denotes the index of head. Following that, a self-attention layer is applied. Finally, the results of all heads are concatenated and passed to a Conv2D followed by a PReLU activation function and a LN layer.

When calculating Q and K , we actually apply compression along the feature dimension when performing convolution. Therefore, a Linear Gate is added to compensate for the information loss during the compression, inspired by SepReformer [30]. Linear Gate consists of an optional Conv2D, a LN layer, a linear layer along frequency dimension, a sigmoid function, and another optional Conv2D. These two Conv2D perform downsampling and upsampling along the frequency dimension, respectively, to prevent the linear layer from having a large number of parameters due to the large frequency dimension. Subsequently, the output of CMHSA Module is multiplied by the output of Linear Gate. Note that the compression in CMHSA Module is performed along the feature dimension, while in Linear Gate it is along the frequency dimension.

Finally, a Conv2D, a GLU and a point-wise Conv2D are applied to obtain the final output. The first Conv2D doubles the feature dimension and feeds the result into the GLU.

4. Experimental Setup

4.1. Dataset and evaluation

We evaluated the performance of Band-SCNet using the MUSDB18-HQ [23], the most popular dataset used in Music

Table 1: Comparison with other models on MUSDB18-HQ. The metrics for All, Drums, Bass, Other and Vocals are SDR (dB). $\times n$ represents that n separate models are required because it is a single-output model. The best scores in real-time (causal) and non-real-time (non-causal) models are highlighted in bold respectively.

Model	Causal	All	Drums	Bass	Other	Vocals	Params (M)	FLOPs (G/s)	Latency (ms)	RTF
HT Demucs [18]	No	7.52	7.94	8.48	5.72	7.93	42.0	-	-	-
Hybrid Demucs [17]	No	7.64	8.12	8.43	5.65	8.35	83.6×3	-	-	-
DTTNet [5]	No	8.06	7.74	7.45	6.92	10.12	5.0×4	-	-	-
BSRNN* [10]	No	8.24	9.01	7.22	6.70	10.01	40.14×3	-	-	-
SCNet [13]	No	9.00	10.51	8.82	6.76	9.89	10.08	-	-	-
SCNet-large [13]	No	9.69	10.98	9.49	7.44	10.86	42.18	-	-	-
BS-RoFormer [11]	No	9.80	9.49	11.31	7.73	10.66	72.2×3	-	-	-
HS-TasNet [4]	Yes	4.65	5.22	4.59	3.64	5.13	42	-	23	-
Wave-U-Net [2]	Yes	6.47	7.05	7.29	4.62	6.91	5.7	-	23	-
Online SCNet	Yes	7.09	8.23	6.16	5.64	8.53	4.36	12.25	92	0.428
Band-SCNet	Yes	7.79	9.44	7.13	5.87	8.74	2.59	17.59	92	0.478

Source Separation. This dataset consists of 100 songs for training and 50 songs for test, with a sampling rate of 44.1 kHz.

In the experiments, we utilize Signal-to-Distortion Ratio (SDR) [31] calculated by museval [1] as the evaluation metric. The number of floating point operations (FLOPs) are calculated by the official tool (torch.utils.flop_counter.FlopCounterMode on meta device). We calculate the FLOPs using 11-second segments. Real-Time Factor (RTF) is used to evaluate the model's efficiency, which is defined as the ratio of the time required to process an input to the duration of the input. We use Intel(R) Core(TM) i7-13700H 2.40 GHz, single-threaded, to test RTF.

4.2. Model configuration

The STFT window size is about 92 ms (4096-point) with a hop size of about 23ms (1024-point), which causes a latency of 92 ms. The configuration of the SD Block is consistent with that of SCNet. The feature dimension $C_i = \{4, 32, 64, 128\}$ for $i = 0, 1, 2, 3$ and the split ratios for low, medium, and high frequencies are 17.5%, 39.2% and 43.3%, respectively. For Separation Network, we make use of the hyperparameters in SpatialNet [20]. The kernel size of T-GConv1d and F-GConv1d are set to 5 and 3, respectively. The number of groups for T-GConv1d, F-GConv1d and GN are all set to 8. The number of self-attention heads is set to 4 and feature dimensions are set to $C' = 8$ and $C'' = 256$. For CSA Fusion Module, we set the number of heads $H = 4$ and the embedding dimension $C_a = 1024$. We only use the optional Conv2D of Linear Gate in CSA Fusion₁, with both the kernel size and stride set to 2.

Following SCNet, we split the music data into overlapped 11-second segments for training and evaluation, and adopt remixing and scaling for data augmentation. We use the root mean squared error loss of complex spectrogram for training.

5. Result

5.1. Results on MUSDB18-HQ

Table 1 compares the proposed Band-SCNet with other previous works on the MUSDB18-HQ dataset. We only explore the conditions using the MUSDB18-HQ dataset without extra training data. Band-SCNet achieves a mean SDR of 7.79 dB with only 2.59M parameters, surpassing Wave-U-Net [2] and HS-TasNet [4] while maintaining fewer parameters, with an accept-

able latency of 92 ms. Compared to Online SCNet, the higher FLOPs of Band-SCNet are due to the use of MHSA, but its RTF remains comparable.

5.2. Ablation experiments

The results of ablation experiments is presented in Table 2. When adjusted to a real-time model, the mean SDR of SCNet drops from 9.00 dB to 7.09 dB, which is a significant gap. Meanwhile, the SDR drop for Drums and Bass is relatively larger, indicating that they are more affected by the limited sequential information. Replacing DPRNN with Cross-band and Narrow-band Blocks increases the mean SDR from 7.09 dB to 7.51 dB, highlighting their effectiveness. And replacing Fusion Modules with CSA Modules increases the SDR from 7.51 dB to 7.79 dB, while reducing parameters and FLOPs, demonstrating the effectiveness of CSA Modules.

Table 2: Results of ablation experiments. "+CNB" represents replacing DPRNN with Cross-band Blocks and Narrow-band Blocks. "+CSA" represents replacing Fusion Modules in SCNet with CSA Fusion Modules, which is equal to Band-SCNet.

Model	All	Drums	Bass	Other	Vocals	Params (M)	FLOPs (G/s)
SCNet	9.00	10.51	8.82	6.76	9.89	10.08	-
Online SCNet	7.09	8.23	6.16	5.64	8.53	4.36	12.25
+CNB	7.51	9.09	6.78	5.66	8.51	2.64	17.7
+CSA	7.79	9.44	7.13	5.87	8.74	2.59	17.59

6. Conclusion

Band-SCNet provides a promising solution for real-time music source separation. By integrating Sparse Compression, Cross-band Blocks, Narrow-band Blocks, and CSA Fusion Modules, it improves performance, reduces model size, and maintains an acceptable latency, confirming its suitability for real-time applications. While the model's latency is relaxed to enhance performance, it still offers an optimal balance between performance and real-time processing. Band-SCNet sets a solid foundation for future advancements in real-time audio processing, with potential for further optimization and application in various real-time audio tasks.

*The parameter size of BSRNN is measured based on the available unofficial code: <https://github.com/amanteur/BandSplitRNN-Pytorch>

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